HEART DISEASE PREDICTION SYSTEM USING MACHINE LEARNING

1. Introduction

Heart disease is one of the leading causes of death worldwide. Early detection of heart disease can help healthcare professionals provide timely treatment and reduce mortality rates. Machine Learning techniques can analyze patient clinical data and predict the likelihood of heart disease with good accuracy.

This project aims to develop a Heart Disease Prediction System using Machine Learning algorithms. The system analyzes patient health parameters and predicts whether a patient is at risk of heart disease. A user-friendly Streamlit interface is also developed to provide real-time predictions.

---

2. Objectives

The main objectives of this project are:

- To analyze heart disease patient data.
- To perform data cleaning and preprocessing.
- To identify important factors affecting heart disease.
- To build machine learning models for prediction.
- To evaluate model performance using standard metrics.
- To develop a patient risk prediction interface.
- To assist in early heart disease detection.

---

3. Dataset Description

The dataset contains clinical information related to heart disease patients.

Features Used

Feature	Description
age	Age of patient
sex	Gender
cp	Chest pain type
Trestbps	Resting blood pressure

Feature	Description
chol	Cholesterol level
fbs	Fasting blood sugar
restecg	Resting ECG results
thalach	Maximum heart rate achieved
exang	Exercise induced angina
oldpeak	ST depression
slope	Slope of ST segment
ca	Number of major vessels
thal	Thalassemia
target	Heart disease status

Target Variable

- 0 = No Heart Disease
- 1 = Heart Disease

---

4. Data Preprocessing

Data preprocessing was performed to improve data quality and model performance.

Steps Performed

- Loaded dataset using Pandas.
- Inspected dataset structure.
- Checked missing values.
- Checked duplicate records.
- Converted data types where required.
- Prepared features and target variables.
- Split dataset into training and testing sets.

Train-Test Split

- Training Data: 80%

- Testing Data: 20%

---

5. Exploratory Data Analysis (EDA)

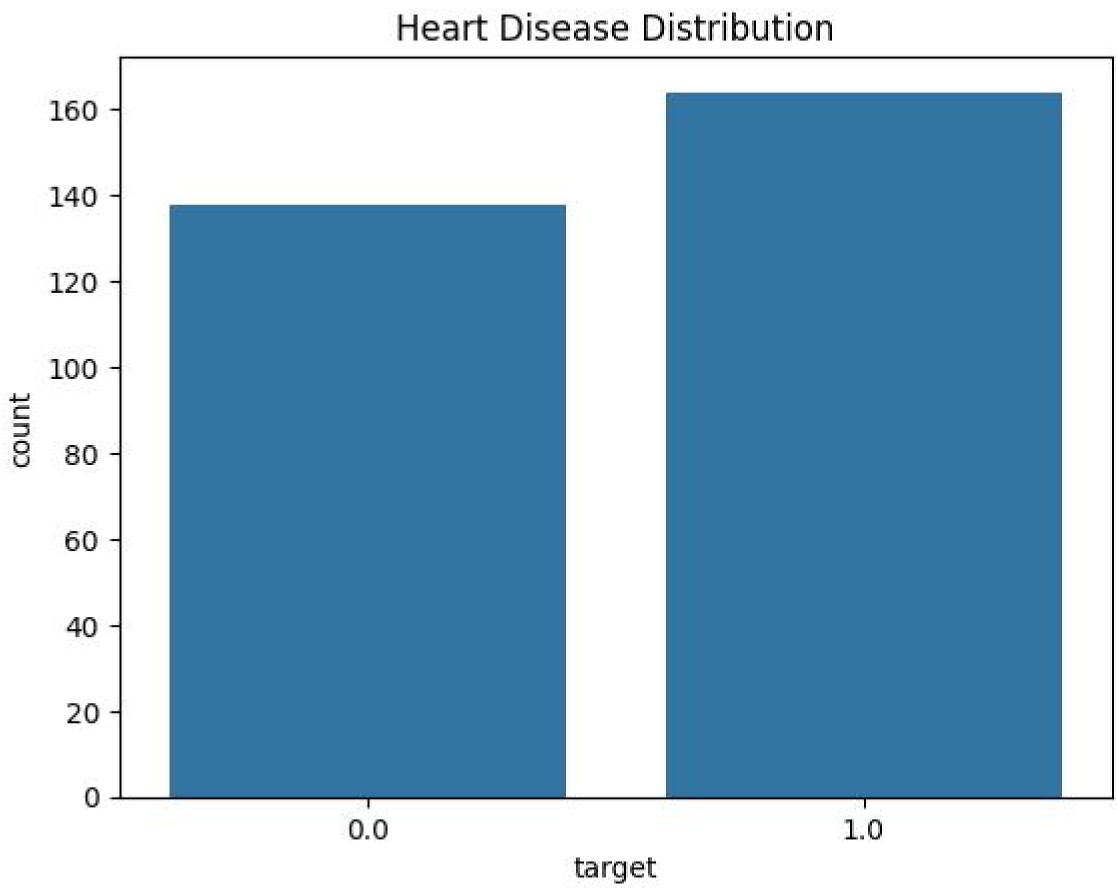
Exploratory Data Analysis was performed to understand the dataset and identify patterns.

Analysis Performed

- Dataset overview
- Target distribution analysis
- Age distribution analysis
- Feature correlation analysis
- Outlier inspection

Observations

- Chest Pain Type has a significant impact on prediction.
- Maximum Heart Rate is strongly associated with heart disease.
- Oldpeak contributes significantly to classification.
- Age shows moderate influence on disease risk.



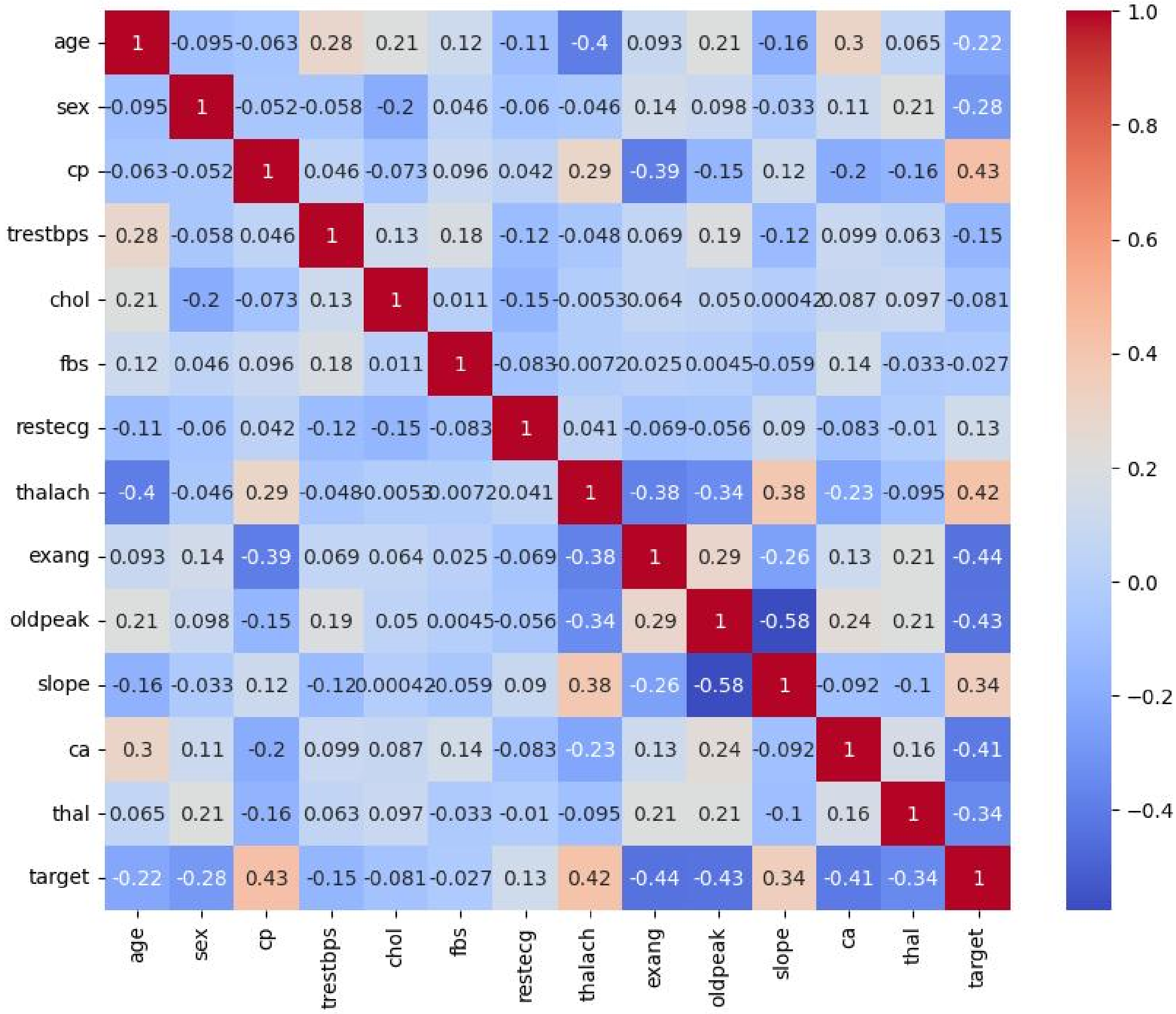
---

6. Correlation Analysis

A correlation heatmap was generated to analyze relationships among features.

Findings

- Several clinical features showed meaningful correlation with the target variable.
- Heart rate, chest pain type, and oldpeak were important predictors.
- Correlation analysis helped understand feature relationships.



7. Machine Learning Models

Two machine learning algorithms were implemented.

7.1 Logistic Regression

Logistic Regression was used as a baseline classification model.

Results

- Accuracy: 80.33%
- ROC-AUC Score: 84.48%

7.2 Random Forest Classifier

Random Forest Classifier was used to improve prediction performance.

Results

- Accuracy: 83.61%
- ROC-AUC Score: 87.88%

Random Forest outperformed Logistic Regression and was selected as the final model.

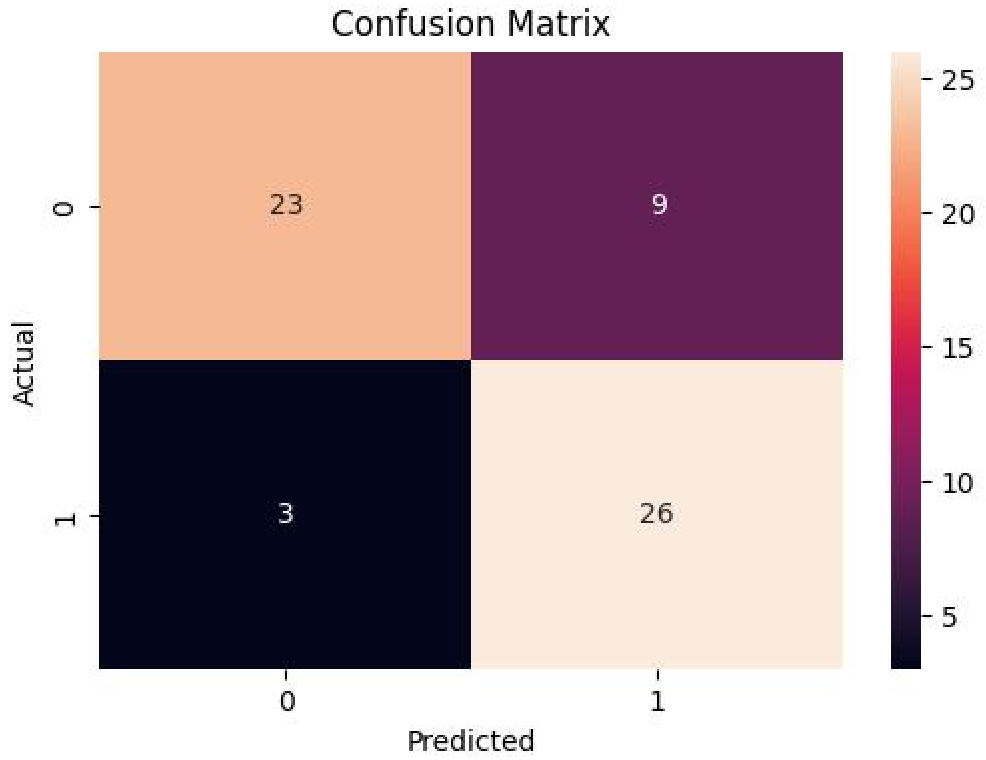
---

8. Model Evaluation

Several evaluation metrics were used.

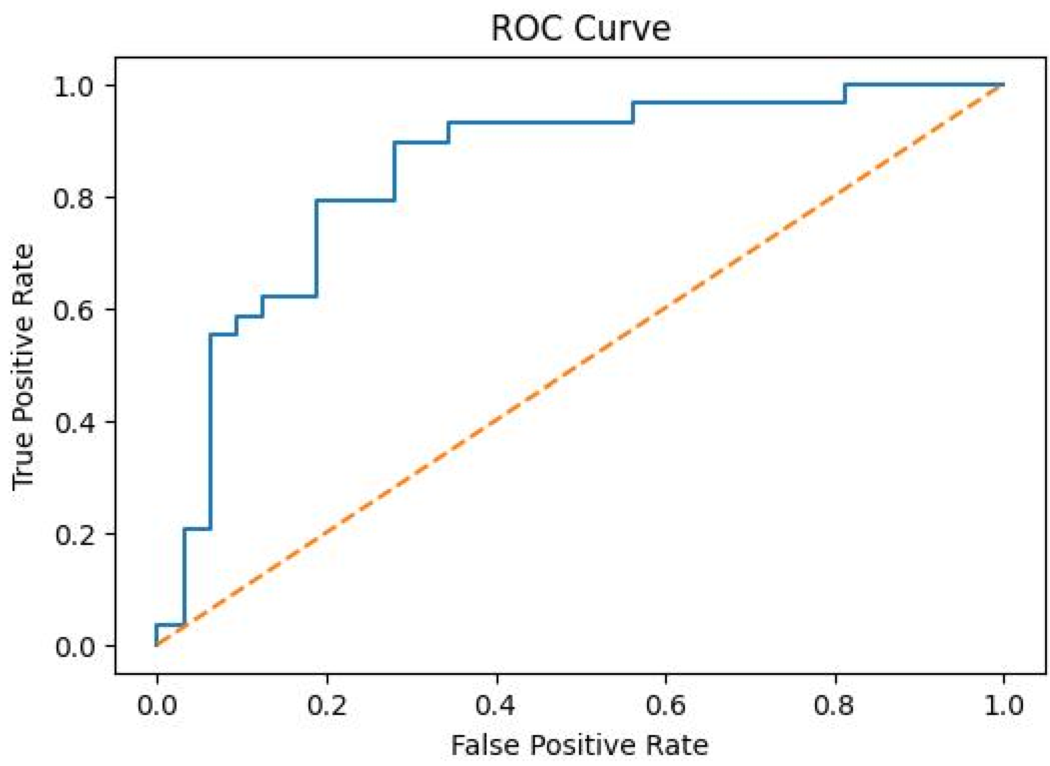
Confusion Matrix

The confusion matrix was generated to evaluate classification performance.



ROC Curve

ROC Curve analysis was performed to measure classification capability.



ROC-AUC Score

Random Forest achieved:

ROC-AUC = 87.88%

This indicates strong classification performance.

---

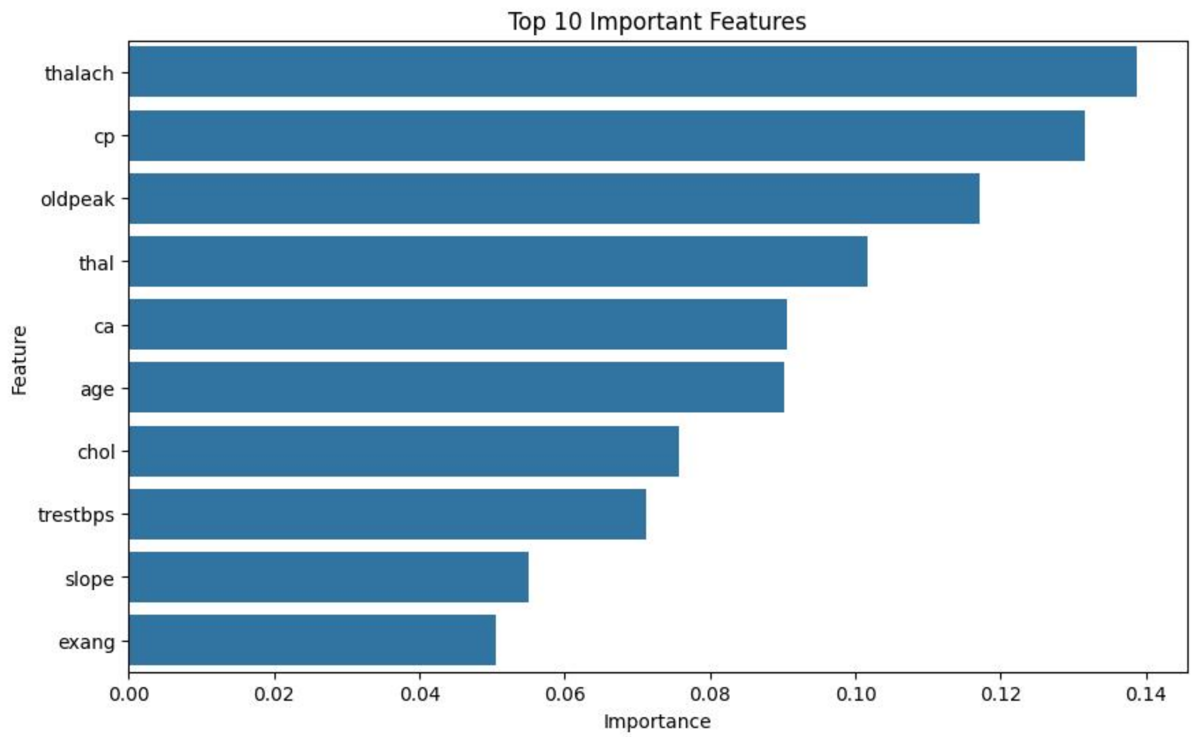
9. Feature Importance Analysis

Feature importance was calculated using the Random Forest model.

Most Important Features

1. thalach
2. cp
3. oldpeak
4. thal
5. ca

These features contributed most to heart disease prediction.



---

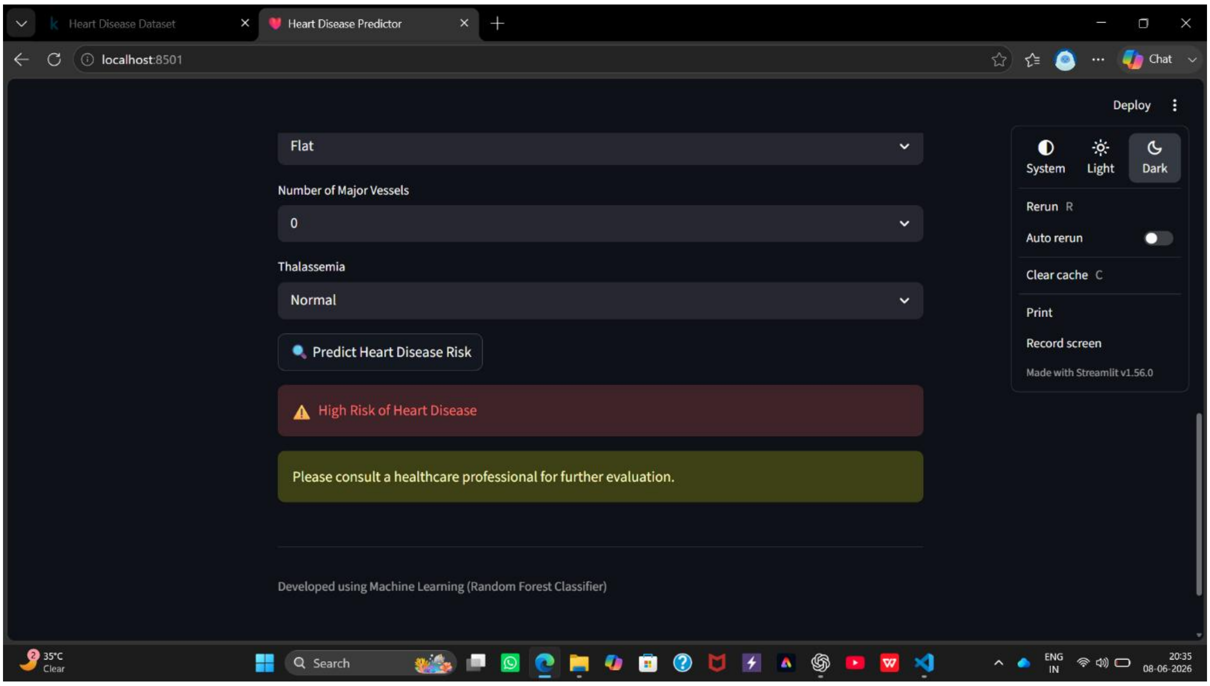
10. Patient Risk Prediction Interface

A web-based prediction system was developed using Streamlit.

Features

- User-friendly interface
- Clinical parameter input
- Real-time prediction
- Risk assessment output

The trained model was exported using Joblib and integrated into the Streamlit application.



---

11. Results Summary

Model	Accuracy	ROC-AUC
Logistic Regression	80.33%	84.48%
Random Forest	83.61%	87.88%

Random Forest achieved the best overall performance.

---

12. Conclusion

A Heart Disease Prediction System was successfully developed using Machine Learning techniques.

The dataset was cleaned, analyzed, and used to train predictive models. Random Forest Classifier achieved the highest performance with an accuracy of 83.61% and a ROC-AUC score of 87.88%.

A Streamlit-based patient risk prediction interface was developed for real-time predictions. The system demonstrates how Machine Learning can support healthcare decision-making and assist in early heart disease detection.

---

13. Future Scope

- Use larger healthcare datasets.
- Integrate deep learning models.
- Connect with hospital databases.
- Deploy application on cloud platforms.
- Improve prediction accuracy using advanced techniques.

References

1. Scikit-Learn Documentation
2. Pandas Documentation
3. Streamlit Documentation
4. Heart Disease Dataset
5. Machine Learning Research Articles